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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

AMMONIUM CHLORIDE, TECHNICAL

The General Service Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description covers technical ammonium chloride in brick form for use in cleaning soldering irons and in white crystal form for use in batteries and for galvanizing purposes.

2. CLASSIFICATION. The technical ammonium chloride shall be of the following types, as specified (see 6.1).

Type I - Ammonium chloride brick.

Type II - Ammonium chloride white crystal.

2.1 Unit weight. The unit weight shall be of the following weights, as specified (6.1).

TABLE I. Ammonium chloride unit weight.

Type I Bricks		Type II White Crystals	
PIN	Weight	PIN	Weight
08	8 ounces (226.8 grams)	08	8 ounces (226.8 grams)
16	16 ounces (453.6 grams)	16	16 ounces (453.6 grams)
32	32 ounces (907.2 grams)	32	32 ounces (907.2 grams)
XX	Other weight as specified	XX	Other weight as specified
NOTE: When XX is included in the part or identification number (PIN) the individual unit weight will be specified by the procuring activity in the purchase order or contract (see 6.2).			

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data that may improve this document should be sent to: Defense Supply Center Richmond, ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 6810

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3. SALIENT CHARACTERISTICS. The ammonium chloride shall conform to the chemical and physical characteristics of table II when tested as specified therein.

TABLE II. Physical and chemical requirements.

Requirements	Minimum or criteria	Maximum or criteria	Test method
Volatile matter, %	-----	1.0	see 3.1
Ammonium chloride content, %	99.0	-----	see 3.2
Residue on ignition (dry basis), %	-----	0.8	see 3.3
Free acid	Pass test	-----	see 3.4
Heavy metals	Pass test	-----	see 3.5
Sieve particle size graduation, %	90.0	-----	see 3.6

3.1 Volatile matter test method. Place approximately 6 grams of the material, weighed to the nearest 0.1 milligram, into a flat-bottomed, tared dish, 2 to 3 inches in diameter, spreading the material over the bottom. Dry over sulfuric acid at room temperature (approximately 25°C) for 4 hours. Weigh to the nearest 0.1 milligram and calculate the percent loss as volatile matter as follows:

$$\text{Percent volatile matter} = \frac{100\,(W-R)}{W}$$

Where: W = weight of specimen used, in grams
R = weight of residue after drying, in grams

3.2 Ammonium chloride content test method. Dissolve approximately 0.2 grams of the dried specimen (see volatile matter), weighted to the nearest 0.1 milligram, in approximately 40 milliliters of water. Add exactly 50 milliliters of 0.1 N silver nitrate solution and 5 milliliters of concentrated nitric acid. Make up the mixture to exactly 100 milliliters with water. Filter through a dry filter and reject the first 10 milliliters of the filtrate. Take exactly 50 milliliters of the filtrate and add 2 milliliters of an 8 percent solution of the ferric ammonium sulfate. Titrate the excess of silver nitrate with 0.1 N ammonium thiocyanate and calculate the percent ammonium chloride as follows:

$$\text{Percent ammonium chloride} = \frac{10.70\,(25N - VM)}{W}$$

Where: N = normality of silver nitrate solution
V = volume of ammonium thiocyanate solution used, in milliliters
M = normality of ammonium thiocynate solution
W = weight of specimen used, in grams

3.3 Residue on ignition (dry basis) test method. Weigh to the nearest 0.1 milligrams approximately 5 grams of the dried specimen (see volatile matter) into a tared crucible. Add

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2 milliliters of concentrated sulfuric acid. Heat at low temperature and then ignite gently to a constant weight. Calculate the percent residue as follows:

$$\text{Percent residue on ignition} = \frac{100\ R}{W}$$

Where: R = weight of residue after ignition, in grams
W = weight of dried specimen used, in grams

3.4 Free acid test method. Dissolve 2 grams of the original specimen in 20 milliliters of water. Add one drop of methyl red indicator. Add one drop of 0.1 N sodium hydroxide. The yellow color shall remain.

3.5 Heavy metals test method. Dissolve 0.5 grams of the original specimen in 50 milliliters of water. Pass hydrogen sulfide gas through the solution and add l milliliter of ammonium hydroxide. No brown color shall develop. Disregard any green color present.

3.6 Sieve particle size graduation test method. The ammonium chloride shall meet the sieve particle size graduation requirement of table I when it is passed through a number 30 sieve in accordance with American Society Testing and Materials (ASTM) E 11, "Wire-Cloth Sieves for Testing Purposes."

4. PRODUCT CONFORMANCE PROVISIONS. The product provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices. It shall be the same as that offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5. PACKAGING. Preservation, packing, and marking shall be as specified in the purchase order or contract.

6. NOTES

6.1 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.

b. Type (see 2) and weight (table I).

c. Quantity required.

d. Packaging requirements.

6.2 Part or identification number (PIN). The following part or identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

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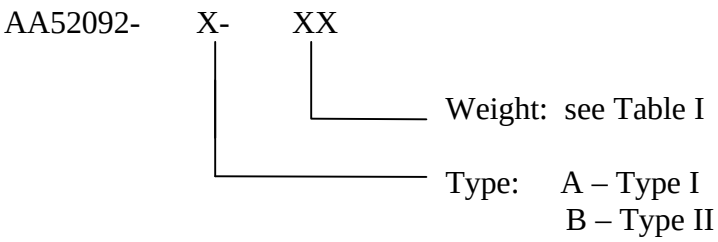

6.3 Material Safety Data Sheets (MSDS). Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets (MSDS) prepared in accordance with FED-STD-313. The pertinent government mailing addresses for submission of data are listed in FED-STD-313. 29 CFR 1910.1200 requires that the MSDS for each hazardous chemical used in an operation must be readily available to personnel using the material. Contracting officers will identify the activities requiring copies of the MSDS.

6.4 Source of documents.

a. American Society for Testing and Materials (ASTM) documents are available from the American Society of Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

b. Government documents are available from the Standardization Document Order Desk, 700 Robbins Ave., Bldg. 4D, Philadelphia, PA 19111-5094.

c. The Code of Federal Regulations (CFR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

MILITARY INTERESTS:

Custodians
Army - GL
Air Force - 68

Reviewers
Army - AR, MD1, SM

CIVIL AGENCY
COORDINATING ACTIVITY:
GSA-FSS

Preparing activity:
DLA-GS

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